



Effect of source and amount of rumen-protected choline on hepatic metabolism during induction of fatty liver in dairy cows

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ABSTRACT

Objectives were to determine the effect of supplementing increased amounts of rumen-protected choline (RPC) from sources with low (L, 28.8%) or high (H, 60.0%) concentration of choline chloride on hepatic metabolism when cows were subjected to feed restriction to develop fatty liver. It was hypothesized that increased supplementation of RPC reduces hepatic triacylglycerol and enhances glycogen concentrations. Pregnant, nonlactating multiparous Holstein cows ($n = 110$) at mean ($\pm$ standard deviation) 232 ± 3.9 d of gestation were blocked by body condition (4.01 ± 0.52) and assigned to receive 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion. Cows were fed for ad libitum intake on d 1 to 5 and restricted to 50% of the NE_L required for maintenance and pregnancy from d 6 to 13. Intake of metabolizable methionine was maintained at 19 g/d during the feed restriction period by supplying rumen-protected methionine. Hepatic tissue was sampled on d 6 and 13 and analyzed for triacylglycerol, glycogen, and mRNA expression of genes involved in choline, glucose, and fatty acids metabolism, cell signaling, inflammation, autophagy, lipid droplet dynamics, lipophagy, and endoplasmic reticulum stress response. Blood was sampled and analyzed for concentrations of fatty acids, β -hydroxybutyrate (BHB), glucose, triacylglycerol, total cholesterol, and haptoglobin. Orthogonal contrasts evaluated the effect of supplementing RPC [CON vs. $(1/4 \cdot L12.9 + 1/4 \cdot L25.8 + 1/4 \cdot H12.9 + 1/4 \cdot H25.8)$], source of RPC [$(1/2 \cdot L12.9 + 1/2 \cdot L25.8)$ vs. $(1/2 \cdot H12.9 + 1/2 \cdot H25.8)$], amount of RPC [$(1/2 \cdot L12.9 + 1/2 \cdot H12.9)$ vs. $(1/2 \cdot L25.8 + 1/2 \cdot H25.8)$], and interaction between source and amount [$(1/2 \cdot L12.9 + 1/2 \cdot H25.8)$ vs. $(1/2 \cdot H12.9 + 1/2 \cdot L25.8)$]. Least squares means and standard error of the means are presented in sequence as CON, L12.9, L25.8, H12.9, H25.8. Supplementation of RPC reduced hepatic triacylglycerol (9.3 vs. 6.6 vs.

5.1 vs. 6.6 vs. $6.0 \pm 0.6\%$ as-is) and increased glycogen contents (1.8 vs. 2.6 vs. 3.6 vs. 3.1 vs. $4.1 \pm 0.2\%$ as-is) on d 13 of the experiment. Feeding RPC reduced serum haptoglobin (136.6 vs. 85.6 vs. 80.6 vs. 82.8 vs. 81.2 ± 4.6 μ g/mL) during the feed restriction period; however, blood concentrations of fatty acids, BHB, glucose, triacylglycerol, and total cholesterol did not differ among treatments. During feed restriction, supplementation of RPC enhanced the mRNA expression of genes related to choline metabolism (*BHMT*), uptake of fatty acids (*CD36*), and autophagy (*ATG3*), and reduced the expression of a transcript associated with endoplasmic reticulum stress response (*ERN1*). An increase in the amount of choline ion from 12.9 to 25.8 g/d enhanced the mRNA expression of genes associated with synthesis and assembly of lipoproteins (*APOB100*), and inflammation (*TNFA*), whereas it reduced the expression of genes linked to gluconeogenesis (*PC*), oxidation of fatty acids (*ACADM*, *MMUT*), ketogenesis (*ACAT1*), and synthesis of antioxidants (*SOD1*) on d 13 of the experiment. Feeding RPC, independent of the product used, promoted lipotropic effects that reduced hepatic lipidosis in dairy cows.

Key words: dairy cow, fatty liver, mRNA, rumen-protected choline

INTRODUCTION

Dairy cows in early lactation mobilize body reserves, mainly adipose but also muscle tissue, as a result of negative nutrient balance (Drackley, 1999). Mobilization of adipose tissue during the periparturient period increases the concentrations of fatty acids in blood, which often leads to the development of fatty liver (Bobe et al., 2004). Arshad and Santos (2022) showed that hepatic lipidosis beyond 4% to 7% triacylglycerol on a wet tissue basis was associated with impaired productive performance, health, and survival in lactating dairy cows. Therefore, interventions such as dietary manipulations to alleviate or prevent hepatic lipidosis are desired because they might not only reduce the risk of fatty liver, but also promote benefits to production and health in dairy cows.

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A dietary intervention to minimize the risk of fat infiltration into the liver is supplying dietary choline as rumen-protected choline (**RPC**; Zenobi et al., 2018a). Choline must be fed to ruminants in a rumen-protected form to reduce microbial degradation in the rumen and increase the supply of choline postrumen (Sharma and Erdman, 1989). Arshad et al. (2020) conducted a meta-analysis with data from 21 randomized experiments to determine the effects of supplemental RPC during the transition period on productive performance and health of parous dairy cows. The authors showed that increasing supplementation of RPC during the transition period linearly increased yields of milk and ECM, with some positive effects on health; nevertheless, they were unable to detect a reduction in hepatic triacylglycerol content in cows supplemented with RPC. In the meta-analysis, only 8 experiments reported concentrations of hepatic triacylglycerol, and in those experiments, cows received 12.8 g/d of supplemental choline ion starting prepartum from the commercial product, except for Elek et al. (2013) who fed 18.6 g/d of choline ion prepartum and 37.6 g/d postpartum from a different source. In the case of Elek et al. (2013), supplementing RPC to transition cows reduced hepatic triacylglycerol content.

A common finding with supplementing RPC is an increase in production without a concurrent increase in DMI (Zenobi et al., 2018b; Bollatti et al., 2020), which could mask the effects of choline on hepatic triacylglycerol (Arshad et al., 2020). Subjecting pregnant dry cows to feed restriction is a model that has been used to evaluate the role of RPC on hepatic metabolism without the potential confounding changes in DMI or nutrients partitioned to milk synthesis. Cooke et al. (2007) conducted 2 experiments using the dry cow feed restriction model and showed that feeding 12.9 g/d of choline ion as RPC alleviated hepatic lipidosis. Zenobi et al. (2018a) showed a linear reduction in hepatic triacylglycerol as the intake of choline ion fed as RPC increased from 0 to 25.8 g/d. Both Cooke et al. (2007) and Zenobi et al. (2018a) supplemented the same RPC product. Choline is a substrate for synthesis of phosphatidylcholine and supplementation of RPC to transition cows increased the mRNA expression of *MTTP* and *APOB100* genes (Goselink et al., 2013), which encode for proteins involved in the biogenesis of lipoproteins such as very-low density lipoprotein (**VLDL**) needed to export triacylglycerol from the hepatic tissue (Bauchart, 1993), and assembly and secretion of **VLDL** require phosphatidylcholine (Yao and Vance, 1988).

We hypothesized that supplementation of choline ion as RPC to cows subjected to feed restriction to induce hepatic lipidosis would reduce triacylglycerol and increase glycogen concentrations in the hepatic tissue and these effects would be linked to changes in expression of

genes involved in hepatic lipid metabolism. Therefore, the objectives were to study the effects of supplementing increased amounts of choline ion from 2 sources of RPC on hepatic tissue composition, blood metabolites, and hepatic mRNA expression of genes involved in pathways linked to choline metabolism, gluconeogenesis, lipid trafficking, assembly and secretion of **VLDL** particles, inflammation, autophagy, lipophagy, lipid droplet dynamics, and endoplasmic reticulum (**ER**) stress response in dairy cows.

MATERIALS AND METHODS

The experiment was conducted at University of Florida Dairy Unit (Hague, FL) from May to October of 2019 and all procedures with cows were approved by the Institutional Animal Care and Use committee of the University of Florida, protocol number 201910612.

Sample Size Calculation

A 2-tailed sample size calculation was performed using the POWER procedure of SAS (version 9.4; SAS/STAT, SAS Institute Inc., Cary, NC). The calculation was based on Zenobi et al. (2018a) in which supplementing 25.8 g/d of choline ion as RPC reduced hepatic triacylglycerol by 6.4 percentage units (SD = 6.0). Because 5 treatments were implemented, the 2-sided sample size was calculated adjusting α to 0.01 by dividing the critical threshold of 0.05 by 5 ($\alpha = 0.05/5$) to allow for multiple comparisons adjustment using the method of Bonferroni and maintain a power of 0.80. Therefore, the sample size calculated ($\alpha = 0.01$; $\beta = 0.20$) resulted in 22 cows per treatment. Because of the experimental design, orthogonal contrasts were pre-planned, and the anticipated power was expected to be greater than 0.80 (difference = 6.4; SD = 6.0) or smaller differences between treatments would be detected.

Experimental Design, Dietary Treatments, Cows, and Housing

The experiment followed a randomized complete block design. Cohorts of 10 multiparous cows were ranked by BCS, and each 5 cows were assigned a block. Within each block, cows were assigned randomly to receive 0, 12.9, or 25.8 g/d of choline ion in choline chloride form (**CC**; ~74.6% choline ion) supplemented with either a low (**L**, 28.8% **CC**; ReaShure, Balchem Corp., New Hampton, NY) or high concentration (**H**, 60.0% **CC**; Prototype, Balchem Corp., New Hampton, NY) in the RPC products. The 5 treatments were **CON**, 0 g/d of choline ion ($n = 22$ cows); **L12.9**, 12.9 g/d of choline ion as RPC (60 g/d ReaShure; $n = 22$ cows); **L25.8**,

25.8 g/d of choline ion as RPC (120 g/d ReaShure; $n = 22$ cows); **H12.9**, 12.9 g/d of choline ion as RPC (28.9 g/d Prototype, $n = 22$ cows); and **H25.8**, 25.8 g/d of choline ion as RPC (57.8 g/d Prototype, $n = 22$). Treatments were supplemented as a top-dress with a mixture of targeted RPC product according to treatment, the coating material containing hydrogenated triacylglycerol to supply similar amounts of lipids from the top-dress in each treatment, and corn meal. Cows assigned to CON received 200 g/d of a top-dress containing the coating material containing hydrogenated triacylglycerol, used to protect CC in the RPC products, and ground corn in a 43:57 ratio. Cows assigned to L12.9, and L25.8 treatments received 200 g/d of a top-dress containing ReaShure, coating material, and ground corn in a 30:21:49, and 60:0:40 ratios, respectively. Cows assigned to H12.9 and H25.8 treatments received 200 g/d of a top-dress containing Prototype, coating material, and ground corn in 14.5:37:48.5 and 29:31:40 ratios, respectively. Of the 110 cows initially enrolled, only 1 cow was removed from the experiment, a cow fed H12.9 and removed on d 6, because it did not acclimate to the research facility and stopped eating any diet or the top-dress. The data from that cow was excluded from all statistical analyses; therefore, 109 cows contributed data to the experiment, 22 CON, 22 L12.9, 22 L25.8, 21 H12.9, and 22 H25.8.

Pregnant, nonlactating parous Holstein cows ($n = 109$) with a mean ($\pm$ SD) 232 ± 3.9 d of gestation and 4.01 ± 0.52 of BCS at enrollment were housed in a cross-ventilated tiestall barn with individual feed bins and water troughs equipped with flow meters. Stalls had air mattresses bedded with sand, and bedding material was cleaned twice daily and replaced thrice a week. The experiment lasted 13 d, with 5 d of ad libitum feeding followed by 8 d of feed restriction.

Ad Libitum Feeding Period

During the ad libitum period (d 1–5), cows were fed a diet designed using the NRC (2001) and it was anticipated that they would consume 10 kg DM/d to achieve a positive NE_L balance. Because of the updated nutrient requirements and supply with NASEM (2021), the amounts of energy and AA supplied by the diets were recalculated using the observed DMI and the nutrient composition of the diets based on the chemical analyses of the ingredients. The recalculated values (NASEM, 2021) resulted in a diet with NE_L content of 1.62 Mcal/kg that resulted in negative energy balance because of the increased energy required for maintenance and the less than anticipated DMI. The supply of MP was 745 g/d with 17 g/d of metabolizable methionine (Table 1).

Cows were fed once daily at 0900 h, and the amounts of feed offered to individual cows were adjusted once daily to ensure at least 5% refusals, which were weighed before the morning feeding. Twice weekly, dietary ingredients were sampled and dried for adjustment of the amounts of DM offered.

Feed Restriction Period

During the feed restriction period, d 6 to 13 of the experiment, the diet was formulated and offered in amounts to supply approximately 50% of the NE_L required for maintenance and pregnancy (NRC, 2001). The mean BW and gestation length of each block of 5 cows was used to calculate the metabolic BW with the associated maintenance requirement and the gestation length, which were used to generate the pregnancy needs using an expected 42 kg of BW for the calf at birth. Based on the NRC (2001), the feed restriction imposed resulted in exactly 50% of the NE_L required; however, the recalculated values using NASEM (2021) resulted in a supply of NE_L of 42% of the required because of the 25% increased maintenance requirement per kg of $BW^{0.75}$ established in NASEM (2021) relative to NRC (2001).

Feed restriction was used to induce lipomobilization and increase accumulation of triacylglycerol into the hepatic tissue as previously reported (Cooke et al., 2007; Zenobi et al., 2018a). The diet was formulated including rumen-protected methionine (Smartamine M, Adisseo USA, Alpharetta, GA) to supply a similar amount of metabolizable methionine as in the ad libitum feeding period (Table 1). The mineral vitamin premix comprised a greater proportion of the diet during the feed restriction period compared with the ad libitum period to supply similar amounts of micronutrients and ionophore (Table 1).

Dietary Ingredient Sampling and Chemical Analyses

Feeds were sampled twice weekly, dried at 55°C for 72 h and at 105°C for 24 h in an air-forced oven, and dry weights were recorded. Samples dried at 55°C were used for chemical analyses, whereas adjustments of the DM offered the DMI of individual cows were calculated based on DM of individual feeds measured at 105°C. Dried individual feed samples were ground to pass a 4-mm screen of a Wiley mill (Thomas Scientific, Swedesboro, NJ). Samples were composited every 2 mo, and 3 composite samples were analyzed using wet chemistry procedures (Dairyland Laboratories Inc., Arcadia, WI). The means and standard deviations are reported in Table 1.

Table 1. Ingredient and nutrient composition of diets fed to pregnant dry cows during the ad libitum and feed restriction periods

Item	Diet ¹	
	Ad libitum	Feed restriction
Ingredient, % of diet DM		
Bermuda grass hay	41.70	38.37
Corn silage	33.30	26.16
Grain mix	25.0	35.47
Composition of grain mix, % of diet DM		
Cottonseed hulls	5.83	13.08
Soybean meal 47%, solvent extract	10.83	11.33
Citrus pulp, dry	5.83	3.49
Prepartum mineral and vitamin mixture ²	2.50	5.23
Vitamin D ³	0.018	0.036
Dicalcium phosphate	—	0.873
Magnesium oxide	—	0.348
Sodium chloride	—	0.700
Rumen-protected methionine ⁴	—	0.383
Nutrient composition, ⁵ DM basis (mean ± SD)		
DM, %	68.9 ± 0.1	73.8 ± 0.5
NE _L , Mcal/kg	1.62	1.50
OM, %	93.3 ± 0.7	90.9 ± 0.2
CP, %	12.5 ± 0.2	14.1 ± 0.5
MP, %	8.83	8.92
MP, g/d	745	442
Methionine, g/d	17	19
Methionine, % MP	2.28	4.30
NFC, %	26.3 ± 3.1	22.4 ± 2.9
NDF, %	51.4 ± 3.4	51.3 ± 2.4
Forage NDF, %	43.7 ± 3.0	38.4 ± 2.7
ADF, %	33.2 ± 0.3	33.1 ± 0.6
Fat, %	2.85 ± 0.3	2.75 ± 0.3
Starch, %	14.1 ± 1.7	12.9 ± 1.8
Ca, %	0.65 ± 0.03	1.09 ± 0.04
P, %	0.25 ± 0.03	0.42 ± 0.02
Mg, %	0.31 ± 0.02	0.67 ± 0.02
K, %	1.26 ± 0.16	1.24 ± 0.12
Na, %	0.28 ± 0.01	0.83 ± 0.02
Cl, %	0.70 ± 0.01	1.52 ± 0.02
S, %	0.19 ± 0.02	0.19 ± 0.02

¹Ad libitum diet was fed from d 1 to 5 for ad libitum intake, whereas the feed restriction diet was fed from d 6 to 13 and offered in amounts to supply 50% of the NE_L required for maintenance and pregnancy (NRC, 2001).

²Contents: 31.0% wheat middlings, 25.2% sodium chloride, 22.6% calcium carbonate, 10.0% magnesium oxide, 7.5% trace mineral and vitamin premix, 0.80% *Saccharomyces cerevisiae* yeast product, and 0.48% Rumensin 90 (Elanco Animal Health, Greenfield, IN). Each kilogram contained 6.2% CP, 12.3% NDF, 1.6% fat, 68.7% ash, 10.95% Ca, 0.42% P, 6.06% Mg, 0.50% K, 0.20% S, 10.31% Na, 15.89% Cl, 1,878 mg of Zn, 272 mg of Cu, 1,655 mg of Mn, 10.2 mg of Se, 26.89 mg of Co, 46.08 mg of I, 97,240 IU of vitamin A, 21,340 IU of vitamin D, 346 IU of vitamin E, and 989 mg of monensin.

³Vitamin D₃ product containing 300 mg of cholecalciferol per kilogram of product (Rovimix D₃, DSM Nutritional Products, Parsippany, NJ).

⁴Smartamine M, 60% metabolizable methionine (Adisseo USA Inc., Alpharetta, GA).

⁵Based on the chemical analyses of dietary ingredients of 3 composite samples for the ad libitum period and 3 composite samples for the feed restriction period. The content of NE_L and supply of MP and methionine were calculated using the NASEM (2021) according to the chemical composition of the dietary ingredients and adjusted for the observed mean DMI of all 109 cows during the ad libitum (8.5 kg/d) and feed restriction (5.0 kg/d) periods. NFC = OM – CP – NDF – ether extract.

BCS and BW

Cows were weighed in the morning before feeding using a walk-through calibrated electronic scale on 2 consecutive days, 1 and 2, and 13 and 14 of the experiment. Values were averaged into single means for the beginning and end of the experiment and the change in

BW calculated. Concurrent with BW, the body condition was scored by a single individual using a 1 to 5 scale with increments of 0.25 units (Ferguson et al., 1994), as depicted in the Elanco BCS chart (Elanco Animal Health, 2009) and the mean value calculated for the beginning and end of the experiment. The change in BCS during the experiment was calculated.

Hepatic Tissue Collection

Hepatic tissue was collected on d 6 and 13 of the experiment. The cow was restrained, and the areas covering the 10th to 12th right intercostal spaces were scanned by ultrasonography (Aloka SSD-500V equipped with a 3.5-MHz convex transducer, Aloka Co. Ltd., Tokyo, Japan) to determine the location for sampling hepatic tissue. The area was shaved and thoroughly disinfected with 3 alternating rounds of povidone-iodine scrubbing followed by a 70% isopropyl alcohol rinse. Upon disinfecting the surgical site, 15 mL of a solution containing 2% lidocaine hydrochloride was administered in the subcutaneous space and intercostal muscles. The surgical site was disinfected again as described previously. After a 2-cm long incision of the skin, a stainless-steel percutaneous hepatic biopsy tool (Aries Surgical, Davis, CA) was introduced aiming for the hepatic tissue visualized previously by ultrasonography and approximately 1 g of tissue was collected. The tissue was placed on filter paper, rinsed with sterile saline, sliced into 3 sections, transferred into 3 separate cryovials, snap-frozen in liquid N, and stored at -80°C until analysis.

Analyses of Hepatic Tissue

The content of triacylglycerol was analyzed on hepatic tissue sampled on d 6 and 13. A block of 5 cows containing all 5 treatments were analyzed in the same assay. Approximately 300 mg of hepatic tissue were processed in triplicate for determination of DM and for extraction of triacylglycerols using chloroform:methanol (Folch et al., 1957). Triacylglycerol was quantified using a colorimetric method (Foster and Dunn, 1973). Details of the procedures for quantification of DM and triacylglycerol in the hepatic tissue are depicted in Arshad and Santos (2022). Intra- and interassay coefficients of variation (CV) were, respectively, 4.1% and 18.5% for hepatic triacylglycerol on a wet basis.

Hepatic glycogen also was analyzed on d 6 and 13. Approximately 100 mg of frozen hepatic tissue was split into 2 specimens and ground to a powder in liquid N_2 using a mortar and pestle and then each 50 mg of tissue was diluted with 500 μL of 2 *M* HCl based on the procedure of Passonneau and Lauderdale (1974). Samples were homogenized with a Precellys 24 Homogenizer (Bertin Instruments, Hialeah, FL) at 5,000 rpm for three 10-s intervals using zirconia beads (2.0 mm, catalog no. 11079124zx, BioSpec Products, Bartlesville, OK). Homogenates were heated at 95°C for 2 h and frequently vortexed to convert glycogen into glucose units. Samples were then centrifuged at $10,000 \times g$ for 10 min at 4°C and the supernatant transferred to a

new tube. One hundred microliters of supernatant was mixed with 300 μL of a 2 *M* solution of HCl and 400 μL of a 2 *M* solution of NaOH. A glucose standard was treated with equal volumes of the same 2 *M* solutions of HCl and NaOH. A reaction buffer contained 100 *mM* triethanolamine, 6.8 *mM* EDTA (pH 7.6), 9.8 *mM* MgCl_2 , 0.5 *mM* NADP, and 1.2 *mM* ATP was added to a series of standards and the unknown samples. Initial absorbance was determined at 340 nm using a microplate reader. Then a cocktail of 1.98 IU of hexokinase and 0.99 IU of glucose-6-phosphate dehydrogenase was added to each well and a second absorbance was determined at 340 nm. The difference between the second and first absorbance readings were determined for the standards and the unknown samples. Using the standard curve, the concentration of glucose per milliliter was calculated for the unknowns and then expressed per unit of hepatic tissue used in the sample (Ramos et al., 2021). Intra- and interassay CV were 5.5% and 5.9%, respectively.

Blood Sampling and Processing

All blood samplings for plasma or serum separation were performed early in the morning before feeding. Blood was sampled from all cows on d 1 (covariate), 6, 7, 9, 11, and 13 by puncture of the coccygeal vessels into 10-mL evacuated tubes (Vacutainer, Becton Dickinson, Franklin Lakes, NJ). For separation of serum, blood was allowed to clot for 30 min, whereas for plasma separation, tubes were placed in ice immediately and transported to the laboratory within 1 h of collection. Tubes were centrifuged for 20 min at $2,000 \times g$ at room temperature for serum or plasma separation. Serum or plasma samples were transferred into multiple aliquots of 1.5 mL and stored frozen at -20°C until analyses.

Blood Measurements

Plasma or serum samples collected on d 1, 6, 7, 9, 11, and 13 of experiment were analyzed for concentrations of fatty acids, BHB, glucose, triacylglycerol, total cholesterol, and haptoglobin. All assays followed the initial randomization with blocks, such that samples from a given block were analyzed in the same assay. Plasma fatty acids (NEFA-C kit; Wako Diagnostics Inc., Richmond, VA) were analyzed according to Johnson and Peters (1993) and plasma BHB (Wako Autokit 3-HB; Wako Diagnostics Inc.) was analyzed using colorimetric enzymatic assays. Intra- and interassay CV were 6.6 and 9.4%, respectively, for fatty acids, and 2.5 and 3.6%, respectively, for BHB. Concentrations of serum glucose were analyzed using a colorimetric assay (Glucose Oxidase G520-480 kit, Teco Diagnostic

Kit, Anaheim, CA). Intra- and interassay CV were 3.9 and 5.2%, respectively. Concentrations of serum triacylglycerol (Stanbio Triglycerides LiquiColor Procedure No. 2100, Stanbio Laboratory, Boerne, TX), and total cholesterol (Stanbio Cholesterol LiquiColor Procedure No. 1010, Stanbio Laboratory) were analyzed using colorimetric enzymatic assays. Intra- and interassay CV were 3.2% and 8.7% for triacylglycerol and total cholesterol 1.7 and 5.2%, respectively. Concentrations of haptoglobin in serum were measured according to Makimura and Suzuki (1982) using a standard curve designed with sera previously analyzed using a commercial kit (Cow Haptoglobin ELISA, Hapt-11; Life Diagnostics Inc., West Chester, PA) from a cow with high (400 µg/mL) and a cow with low (15 µg/mL) concentration of haptoglobin. Intra- and interassay CV were 6.1% and 6.7%, respectively.

Hepatic RNA Extraction and mRNA Expression

Total RNA was extracted from hepatic tissue sampled on d 6 and 13 using Trizol (TRIzol LS Reagent, Invitrogen, Waltham, MA). Approximately 25 mg of tissue was placed in 800 µL of Trizol in a microtube containing zirconium oxide beads (CKMix, Bertin Corp., Thermo Fisher Scientific, Waltham, MA). Chloroform was added to establish a 20% solution. The tissue was homogenized first by hand and then 3 times for 20 s at 6,200 rpm using a homogenizer (Precellys 24, Bertin Corp. Thermo Fisher Scientific). Samples were then centrifuged at $12,000 \times g$ for 15 min at 4°C, and the colorless, aqueous supernatant containing RNA was transferred to a new microtube. Purification of RNA was performed using the Quick-RNA 96 kit (Zymo Research, Irvine, CA) according to the manufacturer's instruction. Purity and concentration were evaluated using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). Samples from d 6 and 13 had a mean ($\pm$ SD) 260:280 nm ratio of 2.02 ± 0.02 , and 2.02 ± 0.02 , respectively (Supplemental Table S1; <https://figshare.com/s/7b0c56a30608a9cb40f0>). Samples from d 6 and 13 had a mean ($\pm$ SD) 260:230 nm ratio of 2.11 ± 0.06 , and 2.12 ± 0.06 , respectively (Supplemental Table S1). The mRNA for a selected set of genes was quantified by the Fluidigm quantitative PCR microfluidic device Biomark HD system (Fluidigm Co., San Francisco, CA). The PCR primers were designed by Fluidigm Delta Gene assays and synthesized by Fluidigm (Fluidigm Co.). Details of genes and primers are presented in Supplemental Table S2 (<https://figshare.com/s/7b0c56a30608a9cb40f0>). A pooled sample containing mRNA from bovine hepatic tissue from 10 different samples was used for primer validation. Primers

validated using Fluidigm primer quality-control criteria ($R^2 \geq 0.97$; efficiency of 90 to 113%; slope = -3.92 to -2.76) were applied to cDNA serially diluted by $12\times$ and evaluated in 8 replicates. Reference genes used were *ACTB*, *GAPDH*, *PGK1*, *RPL19*, *RPS9*, and *YWHAZ*.

The gene panel was constructed to investigate those associated with choline metabolism (*BHMT*, *CEPT1*, *CHKA*, *CHPT1*, *PCYT1A*, *PEMT*, *MTR*, *SLC44A2*, *SLC44A3*), insulin-like growth factor (*IGF1*), growth hormone receptor (*GHR*), gluconeogenesis (*G6PC*, *PC*, *PCK1*, *PCK2*), glycogenesis (*GSK3B*), fatty acids uptake (*CD36*, *SLC27A1*, *SLC27A2*), hepatocellular transport of fatty acids (*FABP1*, *FABP3*, *FABP5*), activation of fatty acids (*ACSL1*, *ACSM1*), carnitine metabolism (*BBOX1*, *CRAT*, *CROT*, *SLC22A5*, *TMLHE*), mitochondrial oxidation of fatty acids (*ACADM*, *CPT1A*, *CPT2*, *HADHA*, *MLYCD*, *MMUT*, *PCCA*), peroxisomal oxidation of fatty acids (*ACADVL*, *ACOX1*, *ECHDC2*, *PPARA*, *PPARG*), re-esterification of fatty acids to triacylglycerol (*ACACA*, *FASN*, *DGAT1*, *DGAT2*), synthesis of phospholipids (*AGPAT2*, *AGPAT3*), cell signaling (*SREBF1*, *SREBF2*, *NFKB1*, *NFKB2*), synthesis and assembly of lipoproteins (*APOB100*, *APOE*, *MTTP*, *PCTP*, *VLDLR*), ketogenesis (*ACAT1*, *HMGCL*, *HMGCS2*), cytokines (*IL1B*, *IL6*, *TNFA*), acute phase response (*HP*, *SAA3*), and synthesis of antioxidants (*GPX3*, *MT1A*, *SOD1*). In addition, the panel also contained genes representing nonclassical pathways associated with hepatic lipidosis (Mashek et al., 2015; Schulze et al., 2017) such as autophagy (*ATG3*, *ATG5*, *ATG7*, *ATG12*, *MAP1LC3B*, *SQSTM1*), lipid droplet dynamics (*PLIN1*, *PLIN2*, *PLIN3*, *PLIN5*), lipophagy (*RAB7A*, *RAB10*, *RAB18*, *RAB32*), and markers of ER stress response (*ATF4*, *ATF6*, *EIF2AK3*, *ERN1*, *GRP78/HSPA5*, *XBP1*).

Primers targeting 5 genes (*FABP3*, *IL6*, *SLC27A1*, *SREBF1*, and *VLDLR*) failed to pass the quality control in the qualification run and were excluded from the analyses. Primer efficiency in the genes passing quality control and used in this experiment ranged between 89.6 and 126.6% (Supplemental Table S2). The geometric mean of the cycle threshold (**Ct**) of all 6 reference genes were calculated for each sample. Statistical analyses were performed on the delta Ct (**dCt**) values as described by Steibel et al. (2009). Fold changes relative to CON were calculated using the method described by Yuan et al. (2006), whereby fold changes were calculated from least squares means (**LSM**) difference or delta-delta Ct (**ddCt**) according to the formula 2^{-ddCt} , where $dCt = Ct_{\text{Target gene}} - \text{geometric mean of } Ct_{\text{Reference genes}}$, and $ddCt = dCt_{\text{Treatment A}} - dCt_{\text{Treatment B}}$. Heatmaps were generated using Heatmapper online tool (Babicki et al., 2016).

Statistical Analyses

Normality of residuals and homogeneity of variance were examined for each continuous dependent variable analyzed after fitting the statistical model. Responses that violated the assumptions of normality were subjected to power transformation according to the BoxCox procedure (Box and Cox, 1964) using a macro (Piepho, 2009) for mixed models in SAS (SAS/STAT, SAS Institute Inc.). Concentrations of haptoglobin in serum had to be transformed before analysis either because of heteroscedasticity or because residuals were not normally distributed. The LSM and standard error of the mean (SEM) were back transformed for presentation of results according to Jørgensen and Pedersen (1998).

Data were analyzed by linear mixed-effects models using the MIXED procedure of SAS (SAS/STAT, SAS Institute Inc.), and analyses were performed separately for the ad libitum and feed restriction periods. The statistical models included the fixed effects of treatment (CON vs. L12.9 vs. L25.8 vs. H12.9 vs. H25.8), the BCS and the BW at the time of enrollment, whether the cow had twins or not (yes or no), the covariate measurement taken before treatment implementation whenever available, and the random effect of block. For responses with repeated measures within cow, the models also included the fixed effects of day of measurement and the interaction between treatment and day, and the random effect of cow nested within treatment. For blood metabolites, the measurement on d 1 was used as a covariate in the statistical analyses. Day was the term in the REPEATED statement. Model fit was assessed with the Akaike's information criterion, and the covariance structure for the models with repeated measures was selected based on the smallest Akaike's information criterion value. In all mixed-effects models, the Kenward-Roger method was used to approximate the denominator degrees of freedom to compute the F tests. When an interaction between treatment and day resulted in $P < 0.10$, then means at different days were partitioned using the SLICE command of SAS (SAS/STAT, SAS Institute Inc.).

In all analyses, single degrees of freedom orthogonal contrasts were evaluated to assess the effects of supplementing RPC [CON vs. $(1/4 \cdot \text{L12.9} + 1/4 \cdot \text{L25.8} + 1/4 \cdot \text{H12.9} + 1/4 \cdot \text{H25.8})$], source of RPC [$(1/2 \cdot \text{L12.9} + 1/2 \cdot \text{L25.8})$ vs. $(1/2 \cdot \text{H12.9} + 1/2 \cdot \text{H25.8})$], amount of RPC [$(1/2 \cdot \text{L12.9} + 1/2 \cdot \text{H12.9})$ vs. $(1/2 \cdot \text{L25.8} + 1/2 \cdot \text{H25.8})$], and interaction between source and amount [$(1/2 \cdot \text{L12.9} + 1/2 \cdot \text{H25.8})$ vs. $(1/2 \cdot \text{H12.9} + 1/2 \cdot \text{L25.8})$]. Data are reported as LSM and SEM.

Evidence against the null hypothesis was considered at $P \leq 0.05$, and tendency was considered at $0.05 < P \leq 0.10$.

RESULTS

A total of 109 cows (22 blocks) contributed data for intake and energy measures, blood metabolites, and hepatic composition, whereas 95 cows (19 blocks) contributed data for hepatic mRNA expression. Among the 109 cows, 9 delivered twins, 2 CON, 1 L12.9, 5 L25.8, and 1 H25.8.

Ad Libitum Feeding Period: Intake of DM and Measures of Energy Status

Feeding RPC or one of the 2 sources of RPC did not affect DMI (Table 2); however, increasing the amount of choline ion from 12.9 to 25.8 g/d reduced ($P = 0.02$) DMI from 8.5 to 7.8 kg/d (Table 1). The reduced DMI with increased amount of choline ion reduced ($P \leq 0.03$) the intake and the balance of NE_L . Treatment did not affect water intake during the ad libitum period.

Ad Libitum Feeding Period: Blood Metabolites

Treatment did not affect the concentrations of fatty acids or BHB in plasma, or the concentrations of total cholesterol or haptoglobin in serum (Table 3; Figure 1). The interaction between source and amount of RPC affected serum glucose and feeding L25.8 reduced glucose concentration compared with feeding L12.9 or H25.8. From enrollment to d 6, concentrations of fatty acids (d 1 = 0.28 vs. d 6 = 0.58 ± 0.05 mM), BHB (d 1 = 0.37 vs. d 6 = 0.55 ± 0.05 mM), and triacylglycerol (d 1 = 14.3 vs. d 6 = 16.0 ± 1.1 mg/dL) increased ($P < 0.01$), whereas concentration of glucose (d 1 = 3.77 vs. d 6 = 3.49 ± 0.12 mM) decreased ($P = 0.01$), and those of total cholesterol and haptoglobin did not change (Figure 1A to 1F). Increasing the amount of choline ion from 12.9 to 25.8 g/d increased ($P = 0.008$) concentrations of triacylglycerol in serum of cows from 15.5 to 17.5 mg/dL (Table 3; Figure 1D).

Ad Libitum Feeding Period: Hepatic Composition

The content of DM in the hepatic tissue tended ($P = 0.07$) to be greater in cows fed 12.9 than 25.8 g/d of choline ion (36.2 vs. 35.2%; Table 4). Treatment did not affect the content of triacylglycerol, glycogen, or the ratio of triacylglycerol to glycogen in the hepatic

Table 2. Effect of source and amount of rumen-protected choline on intake and measures of energy status during the ad libitum and feed restriction periods

Item	Treatment ¹					SEM	P-value ²			
	CON	L12.9	L25.8	H12.9	H25.8		RPC	Source	Amount	Source × amount
BW at enrollment, kg	768	750	764	749	754	22	0.42	0.71	0.54	0.76
BCS at enrollment, 1 to 5	4.01	4.00	4.01	4.05	4.00	0.13	0.96	0.77	0.77	0.65
Ad libitum period ³										
DMI, kg/d	8.00	8.35	7.44	8.71	8.10	0.46	0.68	0.13	0.02	0.64
NE _L intake, Mcal/d	13.0	13.5	12.1	14.1	13.1	0.7	0.68	0.13	0.02	0.64
NE _L balance, Mcal/d	-4.22	-3.76	-5.15	-3.25	-4.18	0.75	0.81	0.17	0.03	0.66
Water intake, L/d	39.7	41.9	34.9	43.6	41.4	4.3	0.86	0.23	0.18	0.47
Feed-restricted period ⁴										
DMI, kg/d	4.93	4.91	4.76	5.09	5.09	0.14	0.72	0.02	0.47	0.47
NE _L intake, Mcal/d	7.39	7.37	7.14	7.64	7.64	0.20	0.72	0.02	0.48	0.46
NE _L balance, Mcal/d	-10.3	-10.5	-10.6	-10.3	-10.2	0.2	0.68	0.12	0.87	0.62
Water intake, L/d	33.8	32.8	33.9	40.4	39.5	3.0	0.26	0.007	0.97	0.67
BW, kg	744	733	740	726	733	15	0.35	0.53	0.54	0.99
BCS, 1 to 5	3.74	3.78	3.78	3.88	3.87	0.10	0.22	0.15	0.97	0.96
Change ⁵										
NE _L intake, Mcal/d	-5.57	-6.17	-4.91	-6.46	-5.48	0.69	0.73	0.38	0.03	0.78
NE _L balance, Mcal/d	-6.06	-6.74	-5.40	-7.03	-6.00	0.69	0.67	0.37	0.02	0.75
BW, kg/d	-1.92	-1.63	-1.98	-1.74	-1.93	0.33	0.72	0.89	0.28	0.74
BCS, 1 to 5	-0.22	-0.23	-0.19	-0.18	-0.13	0.05	0.33	0.13	0.23	0.79

¹Choline chloride (CC) was supplemented as rumen-protected choline (RPC) with either low (L, 28.8% CC) or high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion.

²Orthogonal contrasts: RPC = effect of RPC [CON vs. (1/4-L12.9 + 1/4-L25.8 + 1/4-H12.9 + 1/4-H25.8)]; source = effect of source of RPC [(1/2-L12.9 + 1/2-L25.8) vs. (1/2-H12.9 + 1/2-H25.8)]; amount = effect of amount of RPC [(1/2-L12.9 + 1/2-H12.9) vs. (1/2-L25.8 + 1/2-H25.8)]; source × amount = interaction between source and amount [(1/2-L12.9 + 1/2-H25.8) vs. (1/2-H12.9 + 1/2-L25.8)].

³Cows were fed for ad libitum intake on d 1 to 5 and energy requirements were estimated using NASEM (2021).

⁴Cows were feed restricted to 50% of the NE_L required for maintenance and pregnancy from d 6 to 13 according to NRC (2001), and energy requirements were estimated using NASEM (2021).

⁵Change = value observed during feed restriction – value observed during ad libitum.

tissue on d 6 expressed either on as-is basis or on a DM basis (Table 4).

Ad Libitum Feeding Period: mRNA Expression in Hepatic Tissue

Heatmaps with the genes differentially ($P \leq 0.10$) expressed during the ad libitum period were created using the fold-change relative to CON to illustrate the patterns according to effect of RPC (Figure 2A), and amount of RPC (Figure 2B). The genes affected ($P \leq 0.10$) by treatment during the ad libitum period and the respective fold changes are reported in Table 5, and LSM for the dCt are presented in Supplemental Table S3 (<https://figshare.com/s/7b0c56a30608a9cb40f0>).

Feeding RPC increased ($P < 0.05$) expression of transcripts involved in cellular uptake of fatty acids (*CD36*) and carnitine (*SLC22A5*), cellular autophagy (*ATG5*), biogenesis and stabilization of lipid droplets (*PLIN3*), and ER stress response (*ATF4*), and tended to upregulate a transcript involved in cellular antioxidant protection (*SOD1*; Figure 3A). However, feeding RPC tended ($P = 0.09$) to downregulate the expression of a transcript involved in mitochondrial uptake of propionate (*MMUT*), and downregulated ($P = 0.02$)

a transcript involved in ketogenesis (*HMGCL*; Figure 3A). Feeding RPC had mixed effects on expression of nuclear factor kappa B genes, where it tended ($P = 0.10$) to reduce expression of *NFKB1* but increased ($P = 0.03$) the expression of *NFKB2* (Figure 3A).

Feeding L increased ($P < 0.05$) expression of genes pertaining to regeneration of methionine (*MTR*), activation of long-chain fatty acids (*ACSL1*), carnitine metabolism (*CROT*, *SLC22A5*), mitochondrial oxidation of fatty acids (*HADHA*), re-esterification of fatty acids to triacylglycerol (*DGAT1*), and cellular autophagy (*ATG12*), and tended ($P \leq 0.09$) to upregulate genes involved in peroxisomal oxidation of very-long chain fatty acids (*ACADVL*), and ER stress response (*ATF6*, *HSPA5*, *XBPI1*) compared with H; whereas, expression of a transcript that codes for biogenesis and stabilization of lipid droplets (*PLIN5*) tended ($P = 0.09$) to be upregulated in cows fed H than those fed L.

Increasing the amount of RPC from 12.9 to 25.8 g/d of choline ion increased ($P \leq 0.02$) expression of genes involved in gluconeogenesis (*PCK2*), *HADHA*, synthesis and assembly of VLDL particles (*APOB100*, *MTP*), and *PLIN3*, and tended ($P = 0.07$) to upregulate a transcript involved in hepatocellular transport of fatty acids (*FABP1*). Conversely, increasing

Table 3. Effect of source and amount of rumen-protected choline on blood metabolites during the ad libitum and feed restriction periods

Item	Treatment ¹						P-value ²			
	CON	L12.9	L25.8	H12.9	H25.8	SEM	RPC	Source	Amount	Source × amount
Ad libitum³										
Plasma fatty acids, mM	0.56	0.58	0.59	0.56	0.60	0.05	0.53	0.98	0.47	0.76
Plasma BHB, mM	0.54	0.56	0.55	0.51	0.58	0.05	0.71	0.82	0.41	0.33
Serum glucose, mM	3.25	3.44	3.24	3.32	3.45	0.13	0.23	0.61	0.69	0.04
Serum triacylglycerol, mg/dL	16.5	15.8	18.0	15.1	17.0	1.1	0.99	0.26	0.008	0.87
Serum total cholesterol, mg/dL	135.7	137.4	132.3	133.3	139.8	6.1	0.99	0.73	0.89	0.23
Serum haptoglobin, ⁴ μg/mL	87.0	86.8	85.0	81.5	86.4	7.2	0.70	0.68	0.74	0.47
Feed restricted⁵										
Plasma fatty acids, mM	0.94	0.94	0.93	0.93	0.86	0.06	0.64	0.36	0.35	0.56
Plasma BHB, mM	0.96	0.90	0.87	0.90	0.89	0.06	0.21	0.89	0.63	0.85
Serum glucose, mM	2.99	3.02	3.00	2.96	3.19	0.10	0.48	0.38	0.18	0.10
Serum triacylglycerol, mg/dL	18.6	19.4	20.0	18.1	20.6	1.1	0.30	0.68	0.07	0.23
Serum total cholesterol, mg/dL	125.6	134.4	124.8	129.8	133.9	5.5	0.29	0.60	0.52	0.12
Serum haptoglobin, μg/mL	136.6	85.6	80.6	82.8	81.2	4.6	<0.001	0.85	0.56	0.77

¹Choline chloride (CC) was supplemented as rumen-protected choline (RPC) with either low (L, 28.8% CC) or high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion.

²Orthogonal contrasts: RPC = effect of RPC [CON vs. (1/4-L12.9 + 1/4-L25.8 + 1/4-H12.9 + 1/4-H25.8)]; source = effect of source of RPC [(1/2-L12.9 + 1/2-L25.8) vs. (1/2-H12.9 + 1/2-H25.8)]; amount = effect of amount of RPC [(1/2-L12.9 + 1/2-H12.9) vs. (1/2-L25.8 + 1/2-H25.8)]; source × amount = interaction between source and amount [(1/2-L12.9 + 1/2-H25.8) vs. (1/2-H12.9 + 1/2-L25.8)].

³Cows were fed for ad libitum intake on d 1 to 5 and blood sampled by puncture of coccygeal vessels on the morning of d 6, before imposing feed restriction.

⁴The haptoglobin was transformed to the inverse for statistical analyses and the LSM and respective SEM were back transformed according to Jørgensen and Pedersen (1998).

⁵Cows were feed restricted to 50% of the NE_L required for maintenance and pregnancy from d 6 to 13 according to NRC (2001), and blood was sampled by puncture of coccygeal vessels on d 7, 9, 11, and 13.

the amount of RPC from 12.9 to 25.8 g/d of choline ion decreased ($P \leq 0.05$) the expression of genes associated with peroxisomal oxidation of very-long chain fatty acids (*ECHDC2*) and *NFKB2*, and tended ($P \leq 0.10$) to downregulate genes involved in the synthesis of phosphatidylcholine (*PEMT*), gluconeogenesis (*PCK1*), cellular transporter of fatty acids (*SLC27A2*), and cellular autophagy (*MAP1LC3B*).

A tendency ($P \leq 0.08$) for the interaction between source and amount of RPC was observed for expression of insulin-like growth factor 1 (*IGF1*), acetyl CoA carboxylase (*ACACA*), and a transcript involved in lipophagy (*RAB32*) because feeding L25.8 reduced expression of these transcripts compared with feeding L12.9 or H25.8. A tendency ($P = 0.09$) for the interaction between source and amount of RPC also was observed for a transcript involved in the biosynthesis of carnitine (*TMLHE*) because feeding H12.9 reduced its expression compared with feeding L12.9 or H25.8.

Feed Restriction Period: Intake of DM and Measures of Energy Status

Small differences in intakes of DM, NE_L, and water were observed with source of RPC fed (Table 2). Cows fed L ate 0.26 kg/d less ($P = 0.02$) DM and drank 6.6

L/d less ($P = 0.007$) water than those fed H. Imposing feed restriction resulted in a reduction in NE_L intake of 5.7 Mcal/d, and the decline was 1.1 Mcal/d greater ($P = 0.03$) in cows fed L compared with H. During feed restriction, cows lost a mean of 1.84 kg/d resulting in a loss of BCS of 0.19 units.

Feed Restriction Period: Blood Metabolites

Treatment did not affect the concentrations of fatty acids or BHB in plasma (Table 3; Figures 1A and 1B). A tendency ($P = 0.10$) for the interaction between source and amount of RPC was observed for serum glucose because feeding L25.8 reduced glucose concentration compared with feeding L12.9 or H25.8 (Table 3; Figure 1C). Increasing the amount of choline ion from 12.9 to 25.8 g/d tended to increase ($P = 0.07$) concentrations of triacylglycerol in serum of cows from 18.8 to 20.3 mg/dL (Table 3; Figure 1D). Treatment did not affect the concentrations of total cholesterol in serum (Table 3; Figure 1E). Feeding RPC reduced ($P < 0.001$) the concentrations of haptoglobin in serum from a mean of 136.6 μg/mL in CON to a mean of 82.6 μg/mL in cows fed RPC (Table 3), and the reduction was observed throughout the feed restriction period (Figure 1F).

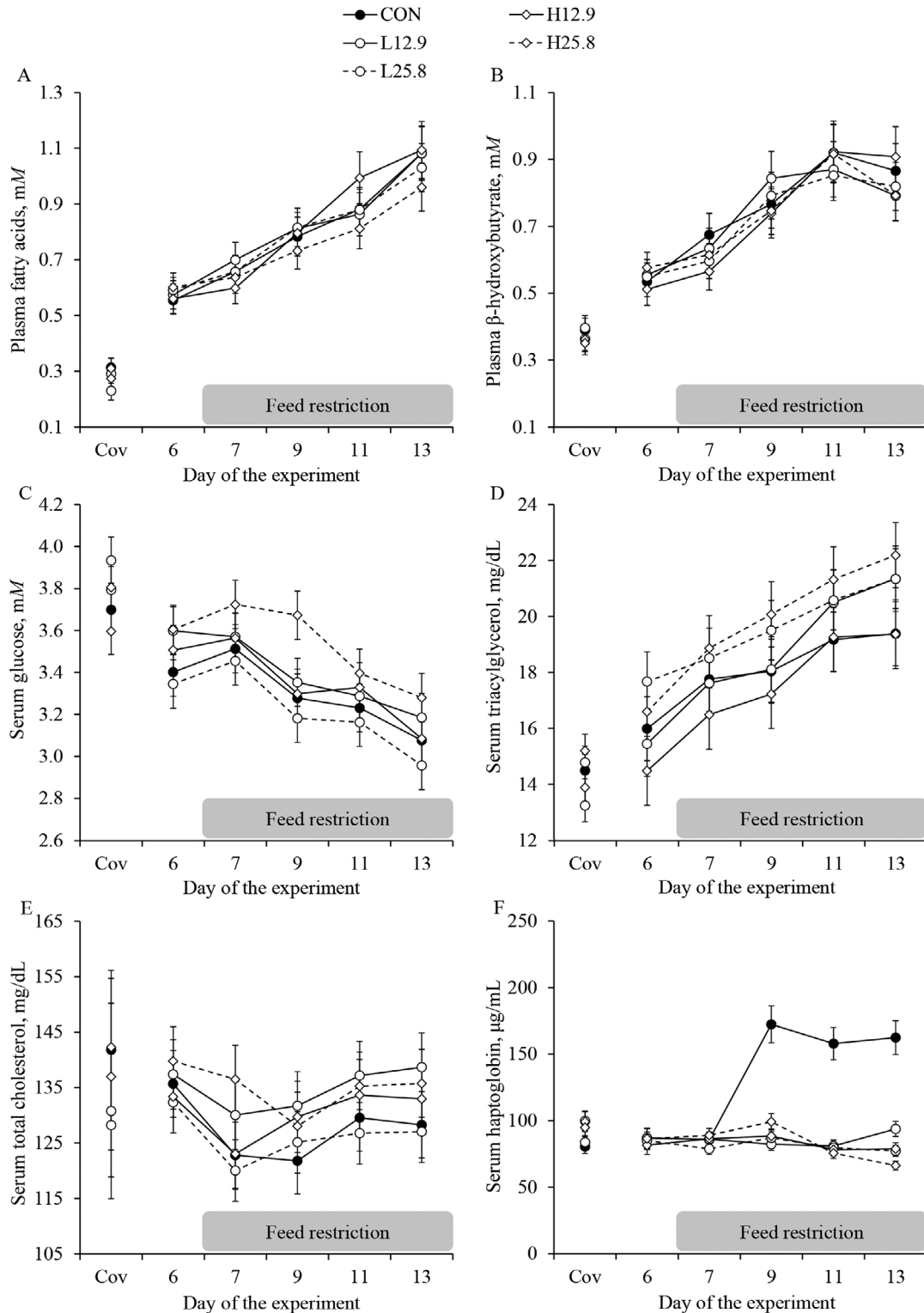


Figure 1. Concentrations of fatty acids (A) and BHB (B) in plasma, and glucose (C), triacylglycerol (D), total cholesterol (E), and haptoglobin (F) in serum during the ad libitum and restricted feeding periods. Choline chloride (CC) was supplemented as rumen-protected choline (RPC) with either low (L, 28.8% CC) or high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion. Panel A: treatment $\times$ day ($P = 0.92$). Panel B: treatment $\times$ day ($P = 0.38$). Panel C: treatment $\times$ day ($P = 0.84$). Panel D: treatment $\times$ day ($P = 0.96$). Panel E: treatment $\times$ day ($P = 0.37$). Panel F: treatment $\times$ day ($P < 0.001$). Cov = covariate value measured immediately before treatment supplementation on d 1 of the experiment. Error bars depict SEM.

Table 4. Effect of source and amount of rumen-protected choline on hepatic composition during the ad libitum and feed restriction periods

Item	Treatment ¹					SEM	P-value ²			
	CON	L12.9	L25.8	H12.9	H25.8		RPC	Source	Amount	Source × amount
Ad libitum ³										
Wet basis										
DM, %	35.6	36.3	35.1	36.0	35.2	1.0	0.89	0.88	0.07	0.72
Triacylglycerol, %	2.45	2.27	2.33	1.90	2.12	0.27	0.21	0.17	0.51	0.70
Glycogen, %	3.53	3.58	3.94	3.78	3.78	0.29	0.30	0.90	0.39	0.38
Triacylglycerol to glycogen, ratio	0.76	0.75	0.72	0.63	0.66	0.10	0.49	0.31	0.99	0.71
DM basis										
Triacylglycerol, %	6.72	6.20	6.51	5.19	5.89	0.72	0.22	0.16	0.38	0.73
Glycogen, %	10.1	10.0	11.2	10.8	10.9	0.9	0.35	0.71	0.33	0.39
Triacylglycerol to glycogen, ratio	0.75	0.75	0.72	0.63	0.66	0.10	0.52	0.30	0.99	0.70
Feed restricted ⁴										
Wet basis										
DM, %	36.0	35.7	34.6	36.5	35.5	1.0	0.53	0.15	0.08	0.98
Triacylglycerol, %	9.32	6.59	5.04	6.61	6.00	0.55	<0.001	0.28	0.02	0.29
Glycogen, %	1.83	2.59	3.55	3.13	4.07	0.18	<0.001	<0.001	<0.001	0.91
Triacylglycerol to glycogen, ratio	5.70	2.62	1.53	2.18	1.55	0.49	<0.001	0.60	0.04	0.57
DM basis										
Triacylglycerol, %	26.3	18.6	14.6	18.3	16.8	1.7	<0.001	0.45	0.03	0.32
Glycogen, %	5.02	7.26	10.31	8.63	11.58	0.57	<0.001	0.002	<0.001	0.90
Triacylglycerol to glycogen, ratio	5.70	2.63	1.53	2.19	1.56	0.49	<0.001	0.61	0.04	0.56

¹Choline chloride (CC) was supplemented as rumen-protected choline (RPC) with either low (L, 28.8% CC) or high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion.

²Orthogonal contrasts: RPC = effect of RPC [CON vs. (1/4-L12.9 + 1/4-L25.8 + 1/4-H12.9 + 1/4-H25.8)]; source = effect of source of RPC [(1/2-L12.9 + 1/2-L25.8) vs. (1/2-H12.9 + 1/2-H25.8)]; amount = effect of amount of RPC [(1/2-L12.9 + 1/2-H12.9) vs. (1/2-L25.8 + 1/2-H25.8)]; source × amount = interaction between source and amount [(1/2-L12.9 + 1/2-H25.8) vs. (1/2-H12.9 + 1/2-L25.8)].

³Cows were fed for ad libitum intake on d 1 to 5 and hepatic tissue collected on the morning of d 6, before imposing feed restriction.

⁴Cows were feed restricted to 50% of the NE_L required for maintenance and pregnancy from d 6 to 13 according to NRC (2001), and hepatic tissue was collected on d 13.

Feed Restriction Period: Hepatic Composition

Feeding RPC reduced ($P < 0.001$) the concentrations of triacylglycerol (CON = 9.3 vs. RPC = $6.1 \pm 0.6\%$ wet basis) and increased ($P < 0.001$) those of glycogen (CON = 1.8 vs. RPC = $3.3 \pm 0.2\%$ wet basis) in the hepatic tissue (Table 4), which resulted in a ratio of triacylglycerol to glycogen that was 2.9-fold smaller ($P < 0.001$) for cows fed RPC compared with CON (CON = 5.69 vs. RPC = 1.97 ± 0.5). The same responses to feeding RPC were observed on hepatic composition analyzed on dry tissue basis (Table 4). Source of RPC ($P < 0.01$) affected the glycogen content of the hepatic tissue whether measured on as-is or DM basis and cows fed H had greater glycogen content than those fed L. Increasing the amount of RPC from 12.9 to 25.8 g/d of choline ion further reduced ($P \leq 0.02$) the concentration of triacylglycerol by 1.1-percentage unit on as-is basis and 2.8-percentage unit on a DM basis (Table 4). Concurrent with a reduction in triacylglycerol, increasing the intake of choline ion from 12.9 to 25.8 g/d increased ($P < 0.001$) the content of hepatic glycogen by 0.9-percentage unit on as-is basis and 3.0-percentage

unit on the tissue DM basis. Such changes in hepatic composition with feeding more RPC resulted in a reduction ($P = 0.04$) in the ratio of triacylglycerol from 2.40 to 1.54.

Feed Restriction Period: mRNA Expression in Hepatic Tissue

Heatmaps with the genes differentially ($P \leq 0.10$) expressed during the feed restriction are depicted according to effect of RPC (Figure 2C), and amount of RPC (Figure 2D). The genes affected ($P \leq 0.10$) by treatment during the feed restriction period and the respective fold changes are reported in Table 6, and LSM for the dCt are presented in Supplemental Table S4 (<https://figshare.com/s/7b0c56a30608a9cb40f0>).

Feeding RPC increased ($P \leq 0.05$) the expression of genes involved in the regeneration of methionine (*BHMT*), cellular autophagy (*ATG3*), and *CD36*, and tended ($P \leq 0.10$) to upregulate the expression of *MTTP*, and genes involved in inflammation (*TNFA*), and acute phase response (*HP*); however, feeding RPC tended ($P = 0.10$) to downregulate the

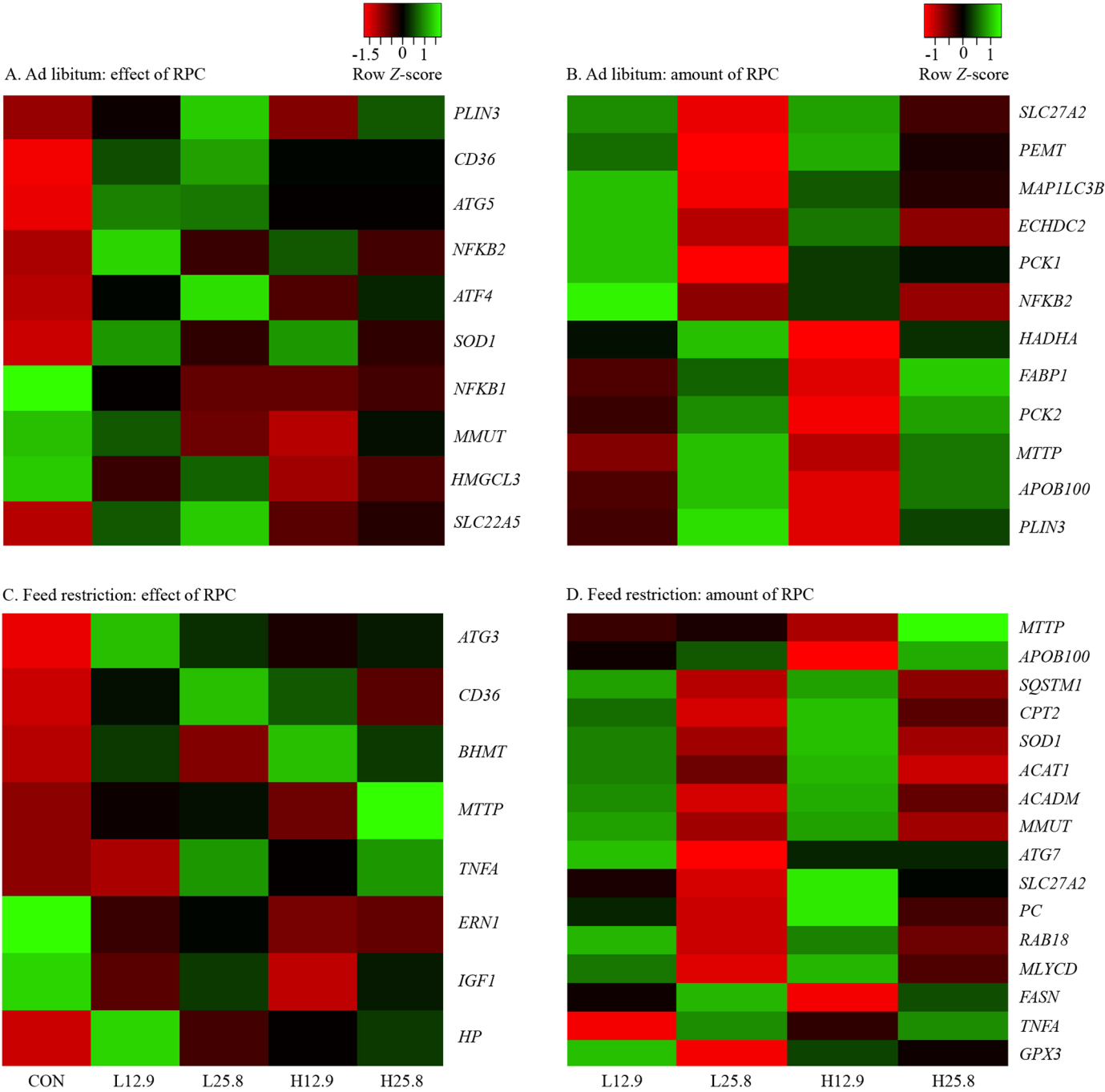


Figure 2. Heat maps including differently ($P \leq 0.10$) expressed genes by rumen-protected choline (RPC) or amount of RPC in cows. Genes were separated using the average linkage clustering method and following the Euclidean distance measurement approach. The mRNA expression increases from red to green, and genes were compared across the rows and presented on a Z-score-based scaling system. Panel A depicts the effect of RPC on mRNA expression in hepatic tissue collected on d 6 of the experiment. Panel B depicts the effect of amount of RPC on mRNA expression in hepatic tissue collected on d 6 of the experiment. Panel C depicts the effect of RPC on mRNA expression in hepatic tissue collected on d 13 of the experiment. Panel D depicts the effect of amount of RPC on mRNA expression in hepatic tissue collected on d 13 of the experiment. Choline chloride (CC) was supplemented as RPC with either a low (L, 28.8% CC) or a high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion.

expression of *IGF1*, and downregulated ($P = 0.02$) a transcript pertaining to ER stress response (*ERN1*; Figure 3B).

Feeding L increased ($P \leq 0.05$) the expression of genes involved in glycogenesis (*GSK3B*) and *DGAT1*, and tended ($P \leq 0.10$) to upregulate genes involved in

Table 5. Effect of source and amount of rumen-protected choline on hepatic relative mRNA expression of genes affected by treatment during the ad libitum period

Item ¹	Treatment ²					P-value ³			
	CON	L12.9	L25.8	H12.9	H25.8	RPC	Source	Amount	Source × amount
Choline metabolism									
<i>MTR</i>	1.00	1.04	1.05	0.93	0.96	0.93	0.04	0.67	0.77
<i>PEMT</i>	1.00	1.09	0.96	1.11	1.04	0.44	0.35	0.08	0.61
Growth factor									
<i>IGF1</i>	1.00	1.00	0.81	0.77	0.96	0.31	0.73	0.96	0.06
Gluconeogenesis									
<i>PCK1</i>	1.00	1.16	0.85	1.07	1.04	0.84	0.52	0.10	0.17
<i>PCK2</i>	1.00	0.99	1.18	0.81	1.20	0.73	0.24	0.001	0.16
Trafficking of fatty acids									
<i>CD36</i>	1.00	1.21	1.26	1.16	1.16	0.01	0.30	0.76	0.76
<i>FABP1</i>	1.00	0.93	1.01	0.86	1.06	0.64	0.82	0.07	0.36
<i>SLC27A2</i>	1.00	1.11	0.96	1.12	1.03	0.47	0.56	0.09	0.63
Activation of fatty acids									
<i>ACSL1</i>	1.00	1.20	1.26	0.78	1.08	0.68	0.03	0.15	0.27
Carnitine metabolism									
<i>CROT</i>	1.00	1.10	1.07	0.77	0.96	0.77	0.03	0.40	0.27
<i>SLC22A5</i>	1.00	1.68	1.96	1.23	1.36	0.001	0.003	0.26	0.80
<i>TMLHE</i>	1.00	1.07	1.01	0.70	0.94	0.46	0.02	0.26	0.09
Mitochondrial oxidation of fatty acids									
<i>HADHA</i>	1.00	1.01	1.12	0.83	1.03	0.89	0.03	0.02	0.39
<i>MMUT</i>	1.00	0.96	0.88	0.85	0.93	0.09	0.54	0.98	0.11
Peroxisomal oxidation of fatty acids									
<i>ACADVL</i>	1.00	1.07	1.15	1.01	1.06	0.14	0.09	0.17	0.79
<i>ECHDC2</i>	1.00	1.18	1.02	1.15	1.04	0.14	0.97	0.04	0.71
Re-esterification of fatty acids									
<i>ACACA</i>	1.00	1.01	0.92	0.87	1.06	0.67	0.94	0.47	0.06
<i>DGAT1</i>	1.00	1.13	1.12	0.76	0.96	0.90	0.04	0.41	0.38
Lipoprotein assembly and synthesis									
<i>APOB100</i>	1.00	1.04	1.26	0.92	1.20	0.29	0.28	0.006	0.67
<i>MTTP</i>	1.00	1.00	1.32	0.95	1.25	0.18	0.48	<0.001	0.96
Ketogenesis									
<i>HMGCL</i>	1.00	0.87	0.95	0.82	0.86	0.02	0.11	0.21	0.71
Cell signaling									
<i>NFKB1</i>	1.00	0.89	0.85	0.85	0.86	0.10	0.83	0.85	0.79
<i>NFKB2</i>	1.00	1.42	1.13	1.28	1.12	0.03	0.51	0.05	0.57
Antioxidant synthesis									
<i>SOD1</i>	1.00	1.09	1.04	1.09	1.04	0.07	0.93	0.12	0.95
Autophagy									
<i>ATG5</i>	1.00	1.36	1.35	1.22	1.22	0.003	0.16	0.99	0.92
<i>ATG12</i>	1.00	1.05	1.02	0.93	0.94	0.70	0.02	0.87	0.63
<i>MAP1LC3B</i>	1.00	1.13	0.96	1.09	1.04	0.39	0.73	0.07	0.33
Lipid droplet dynamics									
<i>PLIN3</i>	1.00	1.13	1.34	1.02	1.23	0.04	0.17	0.01	0.93
<i>PLIN5</i>	1.00	0.95	0.82	1.10	1.13	0.97	0.09	0.66	0.54
Lipophagy									
<i>RAB32</i>	1.00	1.04	0.92	0.85	0.95	0.39	0.23	0.92	0.08
Endoplasmic reticulum stress response									
<i>ATF4</i>	1.00	1.07	1.15	1.04	1.08	0.03	0.19	0.14	0.52
<i>ATF6</i>	1.00	1.02	1.08	0.93	1.00	0.89	0.06	0.14	0.97
<i>HSPA5</i>	1.00	1.13	1.33	0.95	1.10	0.29	0.06	0.13	0.94
<i>XBP1</i>	1.00	1.26	1.21	1.06	1.07	0.13	0.07	0.84	0.74

¹Cows were fed for ad libitum intake on d 1 to 5 and hepatic tissue collected on the morning of d 6, before imposing feed restriction.

²Choline chloride (CC) was supplemented as rumen-protected choline (RPC) with either low (L, 28.8% CC) or high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion.

³Orthogonal contrasts: RPC = effect of RPC [CON vs. (1/4-L12.9 + 1/4-L25.8 + 1/4-H12.9 + 1/4-H25.8)]; source = effect of source of RPC [(1/2-L12.9 + 1/2-L25.8) vs. (1/2-H12.9 + 1/2-H25.8)]; amount = effect of amount of RPC [(1/2-L12.9 + 1/2-H12.9) vs. (1/2-L25.8 + 1/2-H25.8)]; source × amount = interaction between source and amount [(1/2-L12.9 + 1/2-H25.8) vs. (1/2-H12.9 + 1/2-L25.8)].

the biosynthesis of phosphatidylcholine (*CEPT1*) and *MTR* compared with H; whereas, feeding H increased ($P = 0.02$) the expression of *SLC27A2*, and tended ($P = 0.10$) to upregulate *BHMT* compared with L.

Increasing the amount of RPC from 12.9 to 25.8 g/d of choline ion decreased ($P \leq 0.05$) the expression of genes associated with gluconeogenesis (*PC*), mitochondrial oxidation of fatty acids (*ACADM*, *MMUT*), keto-

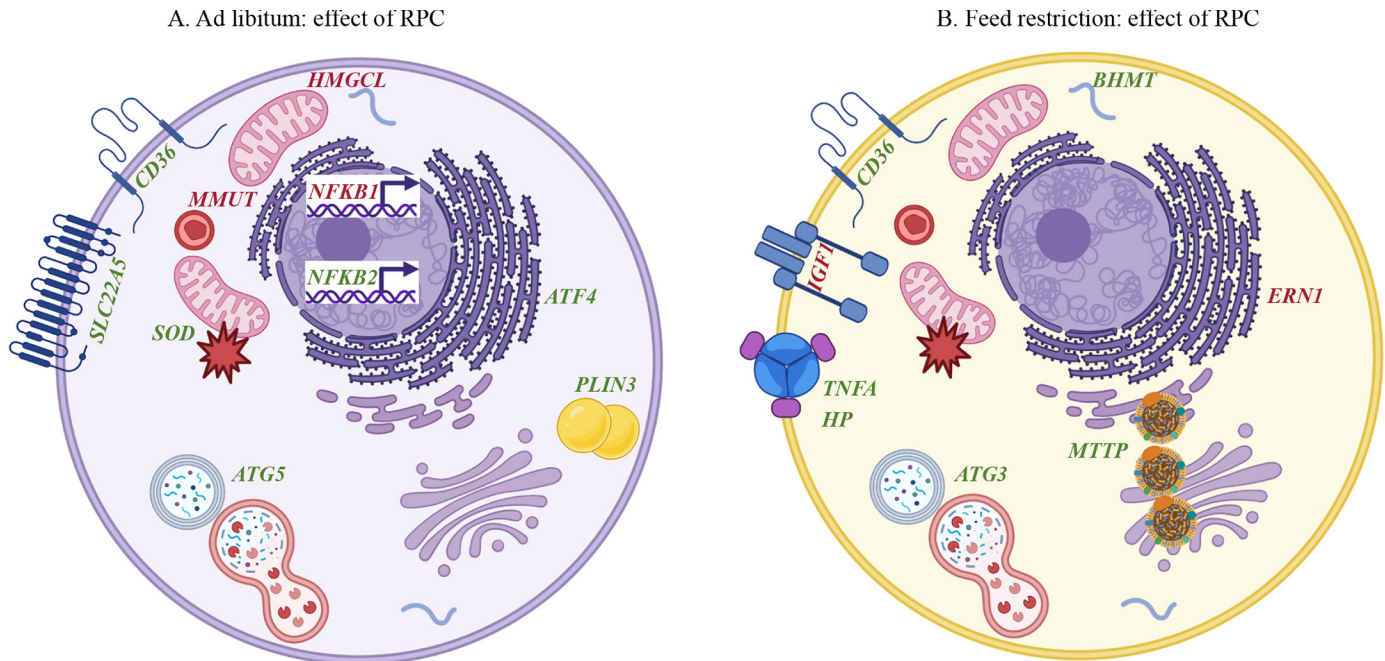


Figure 3. Schematic representations of differently ($P \leq 0.10$) expressed genes by rumen-protected choline (RPC) during the ad libitum (A) or the feed restriction (B) period in dairy cows. Genes are identified associated with the organelles within the hepatocyte. Genes with green color denote upregulation of the transcripts by RPC, whereas genes in red color denote downregulation of the transcripts by RPC.

genesis (*ACAT1*), and cellular antioxidant protection (*SOD1*), and tended ($P \leq 0.10$) to downregulate the expression of *SLC27A2*, and genes involved in mitochondrial oxidation of fatty acids (*CPT2*, *MLYCD*), cellular antioxidant protection (*GPX3*), cellular autophagy (*SQSTM1*), and lipophagy (*RAB18*).

An interaction ($P \leq 0.05$) between source and amount of RPC was observed for genes involved in cellular transporter for choline (*SLC44A3*) and cellular autophagy (*ATG7*) because feeding L25.8 reduced expression of these genes compared with feeding L12.9 or H25.8. A tendency ($P = 0.07$) for the interaction between source and amount of RPC was observed for *MTTP* because feeding H12.9 reduced its expression compared with feeding L12.9 or H25.8.

DISCUSSION

In humans, recommended daily allowances of choline have been established (Institute of Medicine, 1998), and for adults the suggested range is 425 to 550 mg/d (Wallace et al., 2018). Although dietary recommendations of choline have been established for swine, poultry, fish, humans, and dairy calves, no suggested amounts have been recommended for dairy cows (NASEM, 2021). In the present experiment, imposing feed restriction enhanced lipolysis, which was evident based on the increment in concentrations of plasma fatty acids and

BHB, and reduced concentrations of serum glucose, concurrent with a daily loss of approximately 2 kg of BW. The resulting effect was an increase in triacylglycerol content in the hepatic tissue. Nevertheless, feeding RPC markedly reduced the concentration of triacylglycerol and increased those of glycogen in hepatic tissue. The degree of hepatic lipidosis was less as the intake of choline ion increased from 12.9 to 25.8 g/d. Furthermore, the observed changes in hepatic tissue composition were linked with changes in important gene pathways involved in metabolism of choline, lipoprotein synthesis and assembly, cellular autophagy, and ER stress response. The changes in hepatic composition were observed despite lack of differences in blood concentrations of fatty acids thus suggesting that hepatocytes from cows fed RPC were better able to dispose of triacylglycerols resulting in less accumulation into the hepatic tissue.

Effects of RPC on Concentrations of Hepatic Triacylglycerol

Cooke et al. (2007) subjected dry cows to intake of only 30% of the NE_L required for maintenance and pregnancy. Supplementing 12.9 g/d of choline ion as RPC reduced accumulation of triacylglycerol in the liver, although cows fed RPC also had reduced concentrations of fatty acids in plasma (Cooke et al., 2007).

Table 6. Effect of source and amount of rumen-protected choline on hepatic relative mRNA expression of genes affected by treatment during the feed restriction period

Item ¹	Treatment ²					P-value ³			
	CON	L12.9	L25.8	H12.9	H25.8	RPC	Source	Amount	Source × amount
Choline metabolism									
<i>BHMT</i>	1.00	1.19	1.04	1.30	1.19	0.04	0.10	0.12	0.73
<i>CEPT1</i>	1.00	1.06	1.06	0.97	0.99	0.65	0.06	0.89	0.75
<i>MTR</i>	1.00	1.14	1.04	0.98	1.01	0.79	0.10	0.51	0.18
<i>SLC44A3</i>	1.00	1.19	0.97	0.91	1.06	0.77	0.33	0.76	0.04
Growth factor									
<i>IGF1</i>	1.00	0.77	0.88	0.69	0.86	0.10	0.64	0.16	0.70
Gluconeogenesis									
<i>PC</i>	1.00	1.06	0.91	1.18	0.99	0.76	0.21	0.05	0.93
Glycogenesis									
<i>GSK3B</i>	1.00	1.05	1.04	0.97	0.99	0.71	0.04	0.84	0.61
Trafficking of fatty acids									
<i>CD36</i>	1.00	1.12	1.22	1.16	1.06	0.05	0.39	0.94	0.16
<i>SLC27A2</i>	1.00	0.98	0.87	1.13	1.00	0.91	0.02	0.06	0.99
Mitochondrial oxidation of fatty acids									
<i>ACADM</i>	1.00	1.08	0.95	1.09	0.99	0.66	0.62	0.02	0.63
<i>CPT2</i>	1.00	1.07	0.99	1.09	1.02	0.41	0.49	0.09	0.85
<i>MLYCD</i>	1.00	0.98	0.86	1.00	0.91	0.31	0.56	0.10	0.78
<i>MMUT</i>	1.00	1.13	0.99	1.13	0.99	0.28	0.86	0.01	0.91
Re-esterification of fatty acids									
<i>FASN</i>	1.00	0.90	1.09	0.67	0.99	0.51	0.19	0.06	0.54
<i>DGAT1</i>	1.00	1.24	1.07	1.00	1.01	0.35	0.05	0.33	0.29
Lipoprotein synthesis and assembly									
<i>APOB100</i>	1.00	1.08	1.15	0.92	1.21	0.34	0.46	0.04	0.19
<i>MTTP</i>	1.00	1.13	1.16	1.03	1.42	0.09	0.42	0.05	0.07
Ketogenesis									
<i>ACAT1</i>	1.00	1.08	1.00	1.10	0.97	0.49	0.83	0.02	0.50
Cytokines									
<i>TNFA</i>	1.00	0.95	1.49	1.23	1.49	0.07	0.29	0.01	0.27
Acute phase proteins									
<i>HP</i>	1.00	2.17	1.39	1.56	1.72	0.09	0.83	0.56	0.33
Antioxidants synthesis									
<i>GPX3</i>	1.00	2.08	0.89	1.73	1.52	0.16	0.50	0.07	0.17
<i>SOD1</i>	1.00	1.08	0.99	1.10	0.99	0.24	0.67	0.004	0.83
Autophagy									
<i>ATG3</i>	1.00	1.18	1.12	1.09	1.11	0.01	0.31	0.64	0.37
<i>ATG7</i>	1.00	1.17	0.97	1.10	1.10	0.14	0.53	0.06	0.05
<i>SQSTM1</i>	1.00	1.09	1.00	1.09	1.01	0.37	0.95	0.09	0.96
Lipophagy									
<i>RAB18</i>	1.00	1.03	0.90	1.01	0.93	0.53	0.87	0.06	0.60
Endoplasmic reticulum stress response									
<i>ERN1</i>	1.00	0.85	0.88	0.82	0.83	0.02	0.47	0.67	0.85

¹Cows were feed restricted to 50% of the NE_L required for maintenance and pregnancy from d 6 to 13 according to NRC (2001), and hepatic tissue was collected on d 13.

²Choline chloride (CC) was supplemented as rumen-protected choline (RPC) with either low (L, 28.8% CC) or high concentration (H, 60.0% CC) of CC in the RPC product. The amounts supplemented were 0 (CON), 12.9 (L12.9 or H12.9), or 25.8 (L25.8 or H25.8) g/d of choline ion.

³Orthogonal contrasts: RPC = effect of RPC [CON vs. (1/4·L12.9 + 1/4·L25.8 + 1/4·H12.9 + 1/4·H25.8)]; source = effect of source of RPC [(1/2·L12.9 + 1/2·L25.8) vs. (1/2·H12.9 + 1/2·H25.8)]; amount = effect of amount of RPC [(1/2·L12.9 + 1/2·H12.9) vs. (1/2·L25.8 + 1/2·H25.8)]; source × amount = interaction between source and amount [(1/2·L12.9 + 1/2·H25.8) vs. (1/2·H12.9 + 1/2·L25.8)].

Their data implicated reduced lipolysis as one of the potential mechanisms by which feeding RPC might attenuate hepatic lipidosis (Cooke et al., 2007). Nonetheless, others have not been able to document an effect of RPC reducing plasma fatty acids in cows under the same negative energy balance (Zenobi et al., 2018a). Zenobi et al. (2018a) subjected cows to a 9-d feed restriction protocol similar to that of Cooke et al. (2007)

and reported that increasing the intake of choline ion as RPC linearly reduced the concentrations of hepatic triacylglycerol. The changes in hepatic composition were independent of changes in plasma fatty acids or BHB, which are aligned with the findings of the current experiment. Furthermore, Goselink et al. (2013) did not detect any effect of RPC on the expression of genes involved in lipolysis suggesting that lipolytic signals are

not altered by feeding RPC to dairy cows. When cows are in similar nutrient balance, feeding RPC does not seem to affect adipose tissue mobilization or hepatic ketogenesis. Thus, under the conditions of negative nutrient balance, the reduction in hepatic triacylglycerol caused by supplementing choline is likely mediated either by reduced hepatic uptake of fatty acids or increased export of lipoprotein-rich triacylglycerols.

Choline is a precursor for the synthesis of phosphatidylcholine, the predominant phospholipid of VLDL molecules, which are involved in the export of hepatic triacylglycerol. Cows in the last 2 weeks of gestation and the first week of lactation have the smallest concentrations of phosphatidylcholine biomolecules in plasma (Imhasly et al., 2015), coinciding with the period of development of hepatic lipidosis (Bobe et al., 2004). In rats, feeding a choline-deficient diet impaired VLDL secretion by hepatocytes because of phosphatidylcholine deficiency (Yao and Vance, 1988). Supplementing choline to rats increased the synthesis of phosphatidylcholine and the hepatic secretion of VLDL (Yao and Vance, 1988). Thus, it is plausible that supplemental choline may act in a similar fashion in dairy cows, that is, supplying substrate for phosphatidylcholine synthesis that enhances assembly and secretion of VLDL molecules by the liver. Supplementation of choline to bovine hepatocytes in vitro increased VLDL secretion into the culture medium (Chandler and White, 2017). Indeed, recent work by our group showed that cows supplemented with 25.8 g/d of choline ion as RPC had increased hepatic secretion of triacylglycerol-rich lipoprotein when subjected to feed restriction (Arshad et al., 2023). This suggests that reduction in hepatic triacylglycerol in cows supplemented with RPC follows a similar underlying mechanism as that observed in rodents (Yao and Vance, 1988; Rinella et al., 2008). It is noteworthy to mention that during feed restriction, supplementing 25.8 g/d of choline ion as RPC enhanced mRNA expression of *MTTP*, a gene that codes for the large subunit of the heterodimeric MTTP protein. In mice, overexpression of hepatic MTTP protein resulted in increased hepatic secretion of VLDL-triacylglycerol and APOB100 protein (Tietge et al., 1999). Furthermore, increasing the amount of RPC increased expression of *APOB100*, a component of VLDL particles, during feed restriction, which corroborates findings by Goselink et al. (2013) with transition cows. Piepenbrink and Overton (2003) collected hepatic tissue from cows fed increasing amounts of RPC and showed that conversion of carbon 14-labeled palmitate to esterified products decreased within liver slices from cows fed increasing amounts of RPC. In their experiment, oxidation of palmitate by liver slices did not differ by feeding

RPC (Piepenbrink and Overton, 2003). Because hepatic tissue was incubated with palmitate at equivalent concentrations, the observed reduction in palmitate converted to esterified products suggests that feeding RPC increased the rate of triacylglycerol export from hepatocytes. Evidence exists supporting this hypothesis (Arshad et al., 2023). Goselink et al. (2013) fed 12.9 g/d of choline ion to transition cows and showed an increased postpartum hepatic expression of *MTTP* and *APOB100*, both involved in the synthesis and assembly of VLDL in the ER (Olofsson et al., 2000). Sun et al. (2016) showed that concentrations of APOB100 increased in plasma in transition cows fed RPC. Herein, during feed restriction, feeding RPC reduced the expression of *ERN1*, a key sensor for the ER unfolded protein response, and increasing the amount of choline ion from 12.9 to 25.8 g/d increased the expression of *APOB100*, suggesting that RPC might decrease tissue damage and increase proper folding of proteins in the ER compartments of cell to assemble VLDL molecules and, thus, triacylglycerol export from the hepatic tissue.

Effects of RPC on Autophagy

Evidence exists that cows having hepatic triacylglycerol between 1 and 5% on as-is basis exhibited increased expression of autophagy-related genes such as *ATG5*, *ATG7*, *MAP1LC3B*, and *SQSTM1* compared with cows with less than 1% of hepatic triacylglycerol (Chen et al., 2020). Likely, accumulation of hepatic triacylglycerol stimulates the formation and degradation of autophagosomes in hepatic tissue (Chen et al., 2020). In general, autophagy is a cytoprotective conserved process to recycle cell components, which becomes critical for survival during nutrient restriction. Impairment in this adaptive process can lead to accumulation of toxic aggregates, such as misfolded or damaged proteins, or damaged mitochondria that release toxic compounds and reactive oxygen species further increasing the risk of cell death (Ravanan et al., 2017). Supplementation of choline to bovine hepatocytes in vitro reduced accumulation of reactive oxygen species in the culture medium (Chandler and White, 2017). Andrejeva et al. (2020) showed that phosphatidylcholine is required for formation of autophagosome and autolysosomal membranes, whereas phosphatidylethanolamine has an established role in the elongation and assembly of autophagosomal particles (Girardi et al., 2011; Rockenfeller et al., 2015), which can carry out lysosome-mediated degradation of lipid droplets that reduce the risk of nonalcoholic fatty liver (Mashek et al., 2015; Schulze et al., 2017). During feed restriction, supplementation of RPC enhanced

the expression of *BHMT*, an enzyme involved in the transfer of methyl groups from trimethylglycine (i.e., betaine) to homocysteine to regenerate methionine. Increasing expression of *BHMT* by feeding RPC might enhance synthesis of phosphatidylcholine via methylation of phosphatidylethanolamine. Also, feeding RPC during feed restriction increased expression of *ATG3*, involved in autophagy. Thus, it is possible that choline might enhance cellular autophagy to avoid accumulation of toxic aggregates and help mobilize lipid droplets especially in times of greater metabolic activity to alleviate the risk of fatty liver in dairy cows.

Effects of RPC on Concentrations of Hepatic Glycogen

Feeding RPC increased the concentrations of glycogen in the hepatic tissue and increasing choline ion from 12.9 to 25.8 g/d further increased this response. Piepenbrink and Overton (2003) reported a linear increase in the concentration of hepatic glycogen in postpartum dairy cows as intake of choline ion increased from 0 to 16.1 g/d. Likewise, Zenobi et al. (2018a) reported that the concentration of hepatic glycogen tended to increase quadratically with increasing intake of choline ion from 0 to 25.8 g/d in cows subjected to feed restriction. Increased concentration of hepatic glycogen might be because of either reduced glycogenolysis or increased glycogenesis, thus maintaining or more quickly replenishing hepatic glycogen when cows are supplemented with choline. In mice, supplementation of betaine reduced hepatic triacylglycerol and increased glycogen content (Kathirvel et al., 2010). The authors showed that betaine increased tissue sensitivity to insulin based on increased protein abundance of phosphorylated insulin receptor substrate-1 and subsequent downstream effects on PKB/Akt (Kathirvel et al., 2010). The latter are known to phosphorylate GSK3B rendering it inactive, which allows phosphatases to dephosphorylate glycogen synthase to promote glycogen synthesis (Cross et al., 1995; Kathirvel et al., 2010). In the present experiment, RPC did not affect the expression of *GSK3B*, although it is unknown if choline has an effect on hepatic tissue insulin sensitivity in dairy cows that would influence glycogen synthesis. Chandler and White (2019) suggested that methyl groups derived from choline, after oxidation to betaine, are directly tied to the metabolism of glycine and serine. The latter are glucogenic amino acids (Nadkarni et al., 1960), and might be used as substrates for gluconeogenesis and decrease the rate of glycogenolysis. Indeed, serine and choline injection (Munck and Koritz, 1964) and glycine feeding (Hunter et al., 1967) increased the content of hepatic glycogen in rats.

Effects of RPC on Inflammation

Supplementing RPC resulted in smaller ratios of triacylglycerol to glycogen compared with CON. Drackley et al. (1992) suggested that a ratio of hepatic triacylglycerol to glycogen exceeding 1.5 to 2 can be an indicator of susceptibility to fatty liver and clinical ketosis. Increased accumulation of triacylglycerol reduced the ability of hepatocytes to detoxify ammonia to urea (Strang et al., 1998), and convert propionate to glucose (Veenhuizen et al., 1991; Overton et al., 1999) in dairy cows. The ratio of triacylglycerol to glycogen in cows supplemented with RPC was less than half of that of CON cows during feed restriction. Also, feeding RPC reduced concentrations of haptoglobin in serum, an acute phase protein typically stimulated by inflammatory mediators such as TNFA and interleukins (Heinrich et al., 1990). Overfeeding cows prepartum to induce accumulation of hepatic triacylglycerol resulted in increased concentrations of acute phase proteins and TNFA in plasma (Ametaj et al., 2005), suggesting stimulation of the pro-inflammatory response in cows with hepatic lipidosis. In cows fed RPC, the concentration of triacylglycerol in the hepatic tissue increased from d 6 to 13; however, haptoglobin concentrations remained mostly unaltered in those cows. Perhaps, the combined effect of reducing the degree of lipidosis by feeding RPC combined with potential anti-inflammatory effects of phosphatidylcholine (Treede et al., 2007) might have suppressed an increase in haptoglobin in cows fed RPC. Surprisingly, during feed restriction, RPC tended to increase the expression of *HP*, and increasing amount of choline ion from 12.9 to 25.8 g/d increased the expression of *TNFA*, and decreased the expression of *GPX3*, and *SOD1*, and such responses contradict the findings of concentration of haptoglobin in serum; however, concentrations of antioxidants or TNFA proteins were not quantified in serum. Perhaps, posttranscriptional modifications of the *HP* and *TNFA* mRNA via capping, splicing, or polyadenylation altered the translation of the message into protein synthesis, thus creating disagreement between mRNA expression and concentration of haptoglobin in serum. Another possibility is that the turnover of haptoglobin decreased in CON compared with RPC. There is some indication that the half-life of haptoglobin is extended during disease because of reduced turnover (Ohara et al., 1968), which could affect concentrations in serum. In current experiment, supplementing RPC reduced the concentrations of haptoglobin in serum of cows, which corroborates the findings by Zenobi et al. (2018a) in which feeding increased amounts of choline ion linearly reduced the concentrations of haptoglobin in plasma of feed-restricted cows. Furthermore, supplementation

of 12.9 g/d of choline ion to transition cows reduced the concentrations of pro-inflammatory cytokines IL6 and TNFA in blood (Sun et al., 2016), and decreased the expression of *IL1B* and *TNFA*, and TNFA protein production after stimulation of peripheral blood leukocytes (Zenobi et al., 2020). Such responses suggest a potential role of choline to attenuate the inflammatory response in dairy cows.

CONCLUSIONS

Supplementing choline ion as RPC reduced hepatic triacylglycerol and increased glycogen contents without affecting concentrations of fatty acids, BHB, glucose, triacylglycerol, or total cholesterol in blood. The effect of RPC in reducing triacylglycerol and increasing glycogen in the hepatic tissue was enhanced as the amount of choline ion consumed increased from 12.9 to 25.8 g/d. Both sources of RPC were effective in altering hepatic composition during feed restriction. Feeding RPC altered the expression of genes involved in the export of triacylglycerol from hepatic tissue and in disposing lipid droplets in lysosome-mediated autophagy that might help explain the reduced accumulation of triacylglycerol in hepatocytes. Supplementing RPC also reduced the concentrations of haptoglobin in serum, which might have been caused by potential anti-inflammatory effects of lipid-soluble choline-containing phospholipids or by the reduced concentrations of triacylglycerol in the hepatic tissue in cows supplemented with RPC. Collectively, feeding RPC promoted lipotropic effects during negative nutrient balance that reduces the risk of fatty liver in dairy cows. Supplementation of 25.8 g/d of choline ion exhibited more pronounced effects on hepatic metabolism, thus suggesting a dose-dependent effect of choline ion on hepatic responses in dairy cows corroborating previous findings.

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